

PRINCESS JULIANA INTL, Sint Maarten (Dutch part)

WMO: 788660

Lat: 18.046N Lon: 63.115W Elev: 4 StdP: 101.27 Time Zone: -4.00 (W04) Period: 94-19 WBAN: 00399

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.0	22.8	15.8	11.2	24.4	16.8	12.0	24.7	10.0	25.7	9.3	25.7	3.4	90	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.6	32.2	26.4	31.9	26.2	31.2	25.8	27.7	30.9	27.1	30.5	26.7	30.1	4.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.5	30.3	26.1	21.5	29.9	25.8	21.1	29.7	87.8	31.1	85.7	30.6	83.8	30.1	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.0	8.2	7.5	DB	20.8	33.4	1.0	1.7	20.1	34.6	19.5	35.6	18.9	36.5	18.2	37.7	
			WB	17.3	29.0	1.0	1.0	16.6	29.8	15.9	30.4	15.4	30.9	14.6	31.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	25.9	25.8	26.1	26.8	27.7	28.5	28.7	29.0	29.0	28.5	27.5	26.6
	DBStd	1.46	0.78	0.84	0.85	0.85	0.90	0.73	0.82	0.79	0.86	0.93	0.98	0.92
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6393	492	442	499	503	547	556	581	589	569	573	526	514
	CDD18.3	3351	234	209	241	253	289	306	323	331	319	315	276	256
	CDH23.3	34633	1737	1503	1872	2316	3052	3552	3860	4041	3908	3686	2871	2236
(13)	CDH26.7	10226	213	182	289	477	823	1217	1427	1594	1550	1304	763	387
(14)	Wind	WSAvg	4.5	4.8	4.9	4.6	4.5	4.9	5.2	4.8	3.9	3.8	4.1	4.7
(15) Precipitation	PrecAvg	1056	63	47	45	64	99	61	82	102	123	128	139	98
	PrecMax	1802	169	168	169	274	345	190	212	233	363	361	599	305
	PrecMin	704	3	7	7	12	7	3	8	26	26	31	26	15
	PrecStd	253	33	31	31	49	79	42	42	46	69	76	98	60
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	29.2	29.9	30.6	31.5	32.1	32.2	32.8	32.9	32.2	31.2	30.4
		MCWB	24.2	23.8	23.5	25.0	26.0	26.6	26.5	27.0	27.3	26.7	25.7	25.6
	2%	DB	28.9	28.9	29.1	30.0	31.0	31.2	31.9	32.1	32.1	31.7	30.6	29.8
		MCWB	23.9	23.5	23.3	24.6	25.8	25.9	26.1	26.2	26.3	26.3	25.3	24.9
	5%	DB	28.2	28.2	28.8	29.4	30.2	31.0	31.2	31.5	31.8	31.1	30.1	29.1
		MCWB	23.2	23.0	23.2	24.1	25.1	25.7	25.6	25.9	26.1	25.9	25.0	24.2
	10%	DB	28.0	27.9	28.1	29.0	29.9	30.4	30.9	31.1	31.1	30.8	29.8	28.7
		MCWB	23.0	22.8	22.8	23.8	24.8	25.3	25.5	25.8	25.9	25.8	24.9	23.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.8	25.8	25.4	26.2	27.1	27.6	27.9	28.2	28.8	28.2	27.0	26.8
		MCDB	28.6	28.3	28.3	29.5	30.4	31.3	31.4	31.6	31.7	31.1	29.5	29.2
	2%	WB	24.7	24.8	24.6	25.5	26.5	26.9	27.1	27.2	27.3	27.2	26.4	25.7
		MCDB	27.9	27.7	27.9	28.9	29.8	30.5	30.5	30.6	30.9	30.3	29.3	28.5
	5%	WB	24.1	24.0	24.0	25.0	25.9	26.4	26.6	26.7	26.7	26.7	25.9	25.0
		MCDB	27.5	27.3	27.6	28.5	29.3	29.9	30.0	30.1	30.4	29.9	29.0	28.1
	10%	WB	23.5	23.3	23.5	24.4	25.4	25.9	26.1	26.4	26.4	26.2	25.4	24.5
		MCDB	26.9	26.8	27.2	28.1	28.9	29.5	29.6	29.9	30.0	29.6	28.7	27.7
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	4.5	4.6	4.6	4.4	4.3	4.5	4.6	4.6	4.6	4.6	4.5
		MCDBR	4.8	4.7	5.0	5.0	4.7	4.8	4.8	4.9	5.2	4.9	4.8	4.7
		MCWBR	2.8	2.7	2.7	2.7	2.5	2.5	2.4	2.4	2.7	2.6	2.5	2.6
	5% WB	MCDBR	4.7	4.5	4.4	4.5	4.4	4.7	4.7	4.6	4.6	4.5	4.4	4.6
		MCWBR	3.2	3.3	2.8	2.8	2.7	2.9	2.8	2.7	2.8	2.8	2.6	3.0
(40) Clear-Sky Solar Irradiance	taub		0.360	0.368	0.371	0.398	0.439	0.506	0.512	0.475	0.433	0.387	0.371	0.366
	taud		2.613	2.585	2.573	2.518	2.421	2.187	2.163	2.308	2.458	2.603	2.644	2.627
	Ebn at Noon		928	936	939	908	861	797	793	830	867	904	910	911
	Edh at Noon		91	98	102	109	119	148	152	132	113	95	88	88
(44) All-Sky Solar Radiation	RadAvg		4.98	5.72	6.40	6.72	6.51	6.59	6.62	6.57	6.11	5.42	4.79	4.60
(45)	RadStd		0.11	0.11	0.21	0.32	0.37	0.33	0.33	0.22	0.32	0.34	0.28	0.19

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page